Practical 1. Implement the following Simple Servlet applications.

a. Create a simple calculator application using servlet.

Filename : calculator.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Simple Calculator</title>
</head>
<body>
    <h2>Simple Calculator</h2>

    <form action="CalculatorServlet" method="post">
        First Number:
        <input type="number" name="n1" step="any" required>
        <br><br>

        Second Number:
        <input type="number" name="n2" step="any" required>
        <br><br>

        Operation:
        <select name="op">
            <option value="+">Addition</option>
            <option value="-">Subtraction</option>
            <option value="*">Multiplication</option>
            <option value="/">Division</option>
        </select>
```

```
        <br><br>

        <input type="submit" value="Calculate">
    </form>
</body>
</html>
```

CalculatorServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class CalculatorServlet extends HttpServlet {

    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();

        int n1 = Integer.parseInt(request.getParameter("n1"));
        int n2 = Integer.parseInt(request.getParameter("n2"));
        String op = request.getParameter("op");

        int result = 0;
```

```
    if (op.equals("+")) {
        result = n1 + n2;
    } else if (op.equals("-")) {
        result = n1 - n2;
    } else if (op.equals("*")) {
        result = n1 * n2;
    } else if (op.equals("/")) {
        result = n1 / n2;
    }

    out.println("<html>");
    out.println("<body>");
    out.println("<h2>Calculator Result</h2>");
    out.println("<h3>Result: " + result + "</h3>");
    out.println("</body>");
    out.println("</html>");
}

@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    processRequest(request, response);
}

@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    processRequest(request, response);
}
```

```
    @Override
    public String getServletInfo() {
        return "Simple Calculator Servlet";
    }
}
```

b. Create a servlet for a login page. If the username and password are correct then it says message “Hello ” else a message “login failed”

login.html

```
<html>
<head>
    <title>Login Page</title>
</head>
<body>
    <h2>Login Page</h2>

    <form action="LoginServlet" method="post">
        Username:
        <input type="text" name="username" required>
        <br><br>

        Password:
        <input type="password" name="password" required>
        <br><br>

        <input type="submit" value="Login">
    </form>
</body>
</html>
```

LoginServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class LoginServlet extends HttpServlet {

    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();

        String username = request.getParameter("username");
        String password = request.getParameter("password");

        out.println("<html>");
        out.println("<body>");

        if (username.equals("chm") && password.equals("123")) {
            out.println("<h2>Hello " + username + "</h2>");
        } else {
            out.println("<h2>Login Failed</h2>");
        }
```

```
        out.println("</body>");
        out.println("</html>");
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    public String getServletInfo() {
        return "Login Servlet";
    }
}
```

c. Create a registration servlet in Java using JDBC. Accept the details such as Username, Password, Email from the user using HTML Form and store the registration details in the database.

SQL QUERIES

```
CREATE DATABASE studentdb;

USE studentdb;

CREATE TABLE registration (
```

```
    username VARCHAR(50),
    password VARCHAR(50),
    email VARCHAR(100)
);
```

register.html

```
<html>
<head>
    <title>Registration Page</title>
</head>
<body>
    <h2>Registration Form</h2>

    <form action="RegisterServlet" method="post">
        Username:
        <input type="text" name="username" required>
        <br><br>

        Password:
        <input type="password" name="password" required>
        <br><br>

        Email:
        <input type="text" name="email" required>
        <br><br>

        <input type="submit" value="Register">
    </form>
</body>
```

```
</html>
```

RegisterServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class RegisterServlet extends HttpServlet {

    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();

        String username = request.getParameter("username");
        String password = request.getParameter("password");
        String email = request.getParameter("email");

        try {
            Class.forName("com.mysql.cj.jdbc.Driver");

            Connection con = DriverManager.getConnection(
```

```
                "jdbc:mysql://localhost:3306/studentdb",
                "root",
                ""
            );

            PreparedStatement ps = con.prepareStatement(
                "INSERT INTO registration VALUES (?, ?, ?)");

            ps.setString(1, username);
            ps.setString(2, password);
            ps.setString(3, email);
            ps.executeUpdate();
            out.println("<h2>Registration Successful</h2>");
            con.close();
} catch (Exception e) {
        }
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
```

```
    @Override
    public String getServletInfo() {
        return "Registration Servlet";
    }
}
```

Practical 2. Implement the following Servlet applications with Cookies and Sessions.

a. **Using Request Dispatcher Interface create a Servlet which will validate the password entered by the user, if the user has entered "Servlet" as password, then he will be forwarded to Welcome Servlet else the user will stay on the index.html page and an error message will be displayed.**

index.html

```
<html>
<head>
</head>
<body>
   <h2>Enter Password</h2>

   <form action="ValidateServlet" method="post">
      Password:
      <input type="password" name="password" required>
      <br><br>

      <input type="submit" value="Submit">
   </form>
</body>
</html>
```

ValidateServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
```

```
import javax.servlet.RequestDispatcher;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class ValidateServlet extends HttpServlet {

    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");

        String password = request.getParameter("password");

        if (password.equals("Servlet")) {
            RequestDispatcher rd = request.getRequestDispatcher("WelcomeServlet");
            rd.forward(request, response);
        } else {
out.println("Invalid password");
RequestDispatcher rd = request.getRequestDispatcher("ValidateServlet");
            rd.include(request, response);
        }
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
```

```
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    public String getServletInfo() {
        return "Password Validation Servlet";
    }
}
```

WelcomeServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class WelcomeServlet extends HttpServlet {

    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();
```

```
        out.println("<html><body>");
        out.println("<h2>Welcome to Servlet Programming!</h2>");
        out.println("</body></html>");
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    public String getServletInfo() {
        return "Welcome Servlet";
    }
}
```

b. Create a servlet that uses Cookies to store the number of times a user has visited servlet.

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.Cookie;
import javax.servlet.http.HttpServlet;
```

```
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

public class VisitServlet extends HttpServlet {

    protected void processRequest(HttpServletRequest request, HttpServletResponse
response)
        throws ServletException, IOException {

    response.setContentType("text/html;charset=UTF-8");
    PrintWriter out = response.getWriter();

    Cookie[] ck = request.getCookies();
    int count = 0;

    if (cookies != null) {
       for (int i = 0; i < ck.length; i++) {
          if (ck[i].getName().equals("visitCount")) {
             count = Integer.parseInt(ck[i].getValue());
          }
       }
    }

    count++;

    Cookie c = new Cookie("visitCount", count+"");
    c.setMaxAge(60*60*24);
    response.addCookie(c);

    out.println("<h3>You have visited this servlet " + count + " times.</h3>");
```

```
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    public String getServletInfo() {
        return "Visit Counter Servlet";
    }
}
```

c. Create a servlet demonstrating the use of session creation and destruction. Also check whether the user has visited this page first time or has visited earlier also using sessions.

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.http.HttpSession;
```

```
public class SessionServlet extends HttpServlet {

    protected void processRequest(HttpServletRequest request, HttpServletResponse
response)
            throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();

        HttpSession session = request.getSession();
        Integer count = (Integer) session.getAttribute("count");



        if (count == null) {
            count = 1;
            out.println("<h3>You are visiting this page for the first time.</h3>");
        } else {
            count++;
            out.println("<h3>You have visited this page earlier.</h3>");
        }

        session.setAttribute("count", count);

        out.println("<h3>Visit Count: " + count + "</h3>");
        out.println("</body></html>");
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
```

```
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    public String getServletInfo() {
        return "Session Creation Servlet";
    }
}
```

Practical 3. Implement the Servlet IO and File applications.

Develop Simple Servlet Question Answer Application using Database

```
CREATE DATABASE quizdb;

USE quizdb;

CREATE TABLE quiz (
    qno INT PRIMARY KEY,
    question VARCHAR(255),
    op1 VARCHAR(100),
    op2 VARCHAR(100),
    op3 VARCHAR(100),
    op4 VARCHAR(100),
    correctop VARCHAR(100)
);
INSERT INTO quiz VALUES
(1, 'Which language is used for Servlet programming?',
'Python', 'Java', 'C', 'PHP', 'Java');

INSERT INTO quiz VALUES
(2, 'Which method is used to process a POST request?',
'doGet()', 'init()', 'doPost()', 'destroy()', 'doPost()');
```

QuizServlet.java

```
import java.io.*;
import java.sql.*;
import javax.servlet.*;
import javax.servlet.http.*;
```

```
import javax.servlet.annotation.WebServlet;

@WebServlet("/QuizServlet")
public class QuizServlet extends HttpServlet {

    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();

        try {
            Class.forName("com.mysql.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/quizdb", "root", "");

            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("SELECT * FROM quiz");

            out.println("<html><body>");
            out.println("<h2>Online Quiz</h2>");
            out.println("<form action='CheckAnswerServlet' method='post'>");

            while (rs.next()) {
                int qno = rs.getInt("qno");

                out.println("<p>" + qno + ". " + rs.getString("question") + "</p>");
```

```
            for (int i = 1; i <= 4; i++) {
                out.println("<input type='radio' name='q" + qno
                        + "' value='" + rs.getString("op" + i) + "' required>");
                out.println(rs.getString("op" + i) + "<br>");
            }
        }

        out.println("<br><input type='submit' value='Submit Answers'>");
        out.println("</form></body></html>");

        rs.close();
        con.close();

    } catch (Exception e) {
    }
}

@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    processRequest(request, response);
}

@Override
protected void doPost(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {
    processRequest(request, response);
}
}
```

CheckAnswerServlet.java

```
import java.io.*;
import java.sql.*;
import javax.servlet.*;
import javax.servlet.http.*;
import javax.servlet.annotation.WebServlet;

@WebServlet("/CheckAnswerServlet")
public class CheckAnswerServlet extends HttpServlet {

    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();

        try {
            Class.forName("com.mysql.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                    "jdbc:mysql://localhost:3306/quizdb", "root", "");

            Statement st = con.createStatement();
            ResultSet rs = st.executeQuery("SELECT * FROM quiz");

            int score = 0;
            int total = 0;
```

```
            out.println("<html><body>");
            out.println("<h2>Quiz Result</h2>");

            while (rs.next()) {
                total++;

                int qno = rs.getInt("qno");
                String answer = request.getParameter("q" + qno);
                String correct = rs.getString("correctop");

                if (answer != null && answer.equals(correct)) {
                    score++;

                }
            }

            out.println("<h3>Your Score: " + score + " / " + total + "</h3>");
            out.println("</body></html>");

            rs.close();

con.close();

        } catch (Exception e) {

        }
    }
```

```
    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
}
```

PRACTICAL 4. Implement the following JSP applications

a. Develop a simple JSP application to display values obtained from the use of intrinsic objects of various types

implicit.jsp

```
<%@ page import="java.util.Date" %>
<html>
<head>
   <title>JSP Implicit Objects</title>
</head>
<body>
   <h2>JSP Implicit Objects</h2>

   <%
      String name = request.getParameter("name");

      if (name == null) {
         name = "Student";
      }

      session.setAttribute("username", name);
%>

   <p>1. Request Object: <%= name %></p>

   <p>2. Response Object: <%= response.getContentType() %></p>

   <p>3. Out Object: <% out.println("Hello " + name); %></p>
```

```
    <p>4. Session Object: <%= session.getAttribute("username") %></p>


    <p>Current Date: <%= new Date() %></p>
</body>
</html>
```

b. Develop a simple JSP application to pass values from one page to another with validations. (Name-txt, age-txt, hobbies-checkbox, email-txt).

input.jsp

```
<html>
<head>
    <title>Student Details</title>
</head>
<body>
    <h2>Student Details Form</h2>

    <form action="display.jsp" method="post">
        Name:
        <input type="text" name="name" required>
        <br><br>

        Age:
        <input type="number" name="age" min="1" max="100" required>
        <br><br>

        Hobbies:
```

```
        <input type="checkbox" name="hobbies" value="Reading">Reading
        <input type="checkbox" name="hobbies" value="Sports">Sports
        <input type="checkbox" name="hobbies" value="Music">Music
        <br><br>

        Email:
        <input type="email" name="email" required>
        <br><br>

        <input type="submit" value="Submit">
    </form>
</body>
</html>
```

display.jsp

```
<%@ page import="java.util.regex.Pattern" %>
<html>
<head>
    <title>Student Details</title>
</head>
<body>
    <h2>Submitted Details</h2>

    <%

        String name = request.getParameter("name");
        String ageText = request.getParameter("age");
        String email = request.getParameter("email");
        String[] hobbies = request.getParameterValues("hobbies");
```

```
        int age = 0;
        boolean valid = true;

        if (name == null || name.trim().isEmpty()) {
            out.println("Name Error");
        }

        try {
            age = Integer.parseInt(ageText);

            if (age < 1 || age > 100) {
                out.println("Invalid Age");
            }
        } catch (Exception e) {
            valid = false;
        }
if (hobbies == null) {
            out.println("No hobbies selected");
        }

    %>

    </p>
</body>
</html>
```

Create a registration JSP application to register the user based on username and password using JDBC

```
CREATE DATABASE jspdb;
```

```
USE jspdb;

CREATE TABLE users (
    username VARCHAR(50),
    password VARCHAR(50)
);
```

register.jsp

```
<html>
<head>
    <title>User Registration</title>
</head>
<body>
    <h2>Registration Form</h2>

    <form action="save.jsp" method="post">
        Username:
        <input type="text" name="username" required>
        <br><br>

        Password:
        <input type="password" name="password" required>
        <br><br>

        <input type="submit" value="Register">
    </form>
</body>
</html>
```

save.jsp

```
<%@ page import="java.sql.*" %>
<html>
<head>
   <title>Registration Result</title>
</head>
<body>
   <%
      String username = request.getParameter("username");
      String password = request.getParameter("password");

      try {
         Class.forName("com.mysql.jdbc.Driver");

         Connection con = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/jspdb", "root", "");

         PreparedStatement ps = con.prepareStatement(
            "INSERT INTO users VALUES (?, ?)");

         ps.setString(1, username);
         ps.setString(2, password);

ps.executeUpdate();

            out.println("<h2>Registration Successful</h2>");

         con.close();
```

```
        } catch (Exception e) {
            out.println("<h2>Error: " + e.getMessage() + "</h2>");
        }
    %>
</body>
</html>
```

Create a login JSP application to login the user based on username and password using JDBC

login.jsp

```
<html>
<head>
    <title>User Login</title>
</head>
<body>
    <h2>Login Form</h2>

    <form action="check.jsp" method="post">
        Username:
        <input type="text" name="username" required>
        <br><br>

        Password:
        <input type="password" name="password" required>
        <br><br>

        <input type="submit" value="Login">
```

```
    </form>
</body>
</html>
```

check.jsp

```
<%@ page import="java.sql.*" %>
<html>
<head>
    <title>Login Result</title>
</head>
<body>
    <%
        String username = request.getParameter("username");
        String password = request.getParameter("password");

        try {
            Class.forName("com.mysql.jdbc.Driver");

            Connection con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/jspdb", "root", "");

            Statement st = con.createStatement();

            String query = "SELECT * FROM users WHERE username='"
                    + username + "' AND password='" + password + "'";

            ResultSet rs = st.executeQuery(query);

            if (rs.next()) {
```

```
            out.println("<h2>Login Successful</h2>");
            out.println("<h3>Welcome " + username + "</h3>");
        } else {
            out.println("<h2>Login Failed</h2>");
        }

        rs.close();
        st.close();
        con.close();

    } catch (Exception e) {
        out.println("<h3>Error: " + e.getMessage() + "</h3>");
    }
  %>
</body>
</html>
```

PRACTICAL 5. Implement the following intermediate JSP applications

Develop a jsp application to demonstrate use of JSP directives.

Page Directive — page.jsp

```
<%@ page language="java" import="java.util.Date" contentType="text/html" %>

<html>
<head>
    <title>Page Directive</title>
</head>
<body>
    <h2>Page Directive Example</h2>

    <p>Welcome to JSP Programming</p>

    <p>Current Date and Time: <%= new Date() %></p>
</body>
</html>
```

header.jsp

```
<h2>Welcome to My Website</h2>
<hr>
```

Include Directive — main.jsp

```
<%@ page contentType="text/html" %>
<html>
<head>
```

```
    <title>Include Directive</title>
</head>
<body>
    <%@ include file="header.jsp" %>

    <h3>This is the Main JSP Page</h3>
</body>
</html>
```

PRACTICAL 6. Implement the following EJB Applications

Create a Currency Converter application using EJB

CurrencyBean.java

```
import javax.ejb.Stateless;

@Stateless
public class CurrencyBean {

    public double convert(double amount, String type) {

        double result = 0;

        if (type.equals("DTR")) {
            result = amount * 83;
        } else if (type.equals("RTD")) {
            result = amount / 83;
        }

        return result;
    }
}
```

index.html

```
<html>
<head>
    <title>Currency Converter</title>
</head>
```

```
<body>
    <h2>Currency Converter</h2>

    <form action="CurrencyServlet" method="post">
        Enter Amount:
        <input type="number" name="amount" step="any" min="0" required>
        <br><br>

        Select Conversion:
        <select name="type">
            <option value="DTR">Dollar to Rupee</option>
            <option value="RTD">Rupee to Dollar</option>
        </select>
        <br><br>

        <input type="submit" value="Convert">
    </form>
</body>
</html>
```

CurrencyServlet.java

```
import java.io.IOException;
import java.io.PrintWriter;
import javax.ejb.EJB;
import javax.servlet.ServletException;
import javax.servlet.annotation.WebServlet;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
```

```
@WebServlet("/CurrencyServlet")
public class CurrencyServlet extends HttpServlet {

    @EJB
    CurrencyBean bean;

    protected void processRequest(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html;charset=UTF-8");
        PrintWriter out = response.getWriter();

        double amount = Double.parseDouble(request.getParameter("amount"));
        String type = request.getParameter("type");

        double result = bean.convert(amount, type);

        out.println("<html><body>");
        out.println("<h2>Currency Conversion Result</h2>");

        if (type.equals("DTR")) {
            out.println("<h3>Dollar to Rupee: ₹" + result + "</h3>");
        } else {
            out.println("<h3>Rupee to Dollar: $" + result + "</h3>");
        }

        out.println("<a href='index.html'>Go Back</a>");
        out.println("</body></html>");
```

```
    }

    @Override
    protected void doGet(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }

    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        processRequest(request, response);
    }
}
```